

Sajjad “JJ” Arshad

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SUMMARY

- **10+ years of software development** experience across multiple programming languages (e.g., C/C++, Python, Java, Go, Bash, NodeJS), with a strong foundation in **data structures and algorithms**.
- Demonstrated cross-org coordination skills by leading numerous **cross-team/cross-functional** working groups to develop strategy roadmaps and deliver significant, complex solutions.
- Expert in conducting large-scale **Web security and privacy** measurements, specifically focused on evaluating the effectiveness of Web attacks and analyzing the dynamics of the online advertising ecosystem.
- Proficient in **advanced Linux programming**, including process management, IPC (signals, shared memory, semaphores, sockets), and multi-threading, with a strong background in Linux internals.
- Developing efficient **large-scale static analysis** algorithms to detect harmful behaviors in cross-platform frameworks within **Android** apps, focusing on languages like Dalvik Bytecode, JavaScript, and native code.
- Expanded my expertise in **GenAI/LLM** by exploring concepts such as **RAG, ReAct, Embeddings, and Agents**, while also gaining practical experience with Gemini API and SAX.
- Possessed a strong understanding of **networking** (e.g., TCP/IP, UDP, Raw socket, Switching, Routing), **security protocols** (e.g., TLS/SSL, PKI, Firewall, IDS/IPS, WAF), and **containerization** (e.g., Docker, Kubernetes).
- Published over **15 papers in security conferences** like USENIX Security, NDSS, and WWW, with Web Cache Deception (WCD) research recognized as [PortSwigger's Top Web Hacking Technique of 2019](#).
- Extensive hands-on experience in **offensive security**, with expertise in system security techniques (e.g., shellcode, ROP, DEP, ASLR, heap exploitation, sandboxing, CFI) and tooling (e.g., Ghidra, IDA Pro, Binary Ninja). Proficient in web security, covering topics such as XSS, CSRF, SSRF, Web Cache Deception, OAuth, and CORS. Decent background in memory safe system languages such as Rust.
- Actively contribute to the cybersecurity community by sharing my expertise through **Android app hacking** workshops at prominent events like [DEFCON](#), [BSidesSF](#), [HackerOne](#), and [H@cktivityCon](#), while also organizing hacking competition like [GoogleCTF](#) and ensuring the preservation of knowledge through maintaining CTF [archives](#) and [writeups](#).

EXPERIENCE

SENIOR SOFTWARE ENGINEER @ Google, Mountain View, CA

May 2019 - Present

- Leading development of **static analysis & RE tooling** (e.g. Semgrep) to detect Android abuse at scale.
- Owning multiple **backend RPC services** responsible for large-scale abuse detection in Google Play.
- Our Android real-time malware blocking technology received widespread press coverage appearing in outlets such as [Bleeping Computer](#), [Tech Radar](#), [Ars Technica](#), [9to5Google](#), and [The Strait Times](#).
- Mastered **Google's CI/CD** toolkits, data storage platforms (e.g., Bigtable, Spanner), microservice frameworks (e.g., gRPC, Protobuf), and load balancing infra to deploy large-scale distributed systems.
- Built a decompiler that allows for the recovery of JavaScript source code from **React Native's Hermes** bytecode.
- Enhanced our Android app review platform by incorporating **LLM**, enabling it to automatically generate concise summaries and offer improved code comprehension tools for the reviewers.

RESEARCH INTERN @ Mozilla Corporation, Mountain View, CA

June 2017 - August 2017

- Conducted a measurement study on the adoption of **TLS 1.3** by middleboxes in enterprise networks.
- Developed a Firefox add-on to collect data from Firefox users into Telemetry platform (Spark) for offline analysis.
- Enhanced Firefox browser code base (C++) to enable flexible HTTPS connection configuration for developers.

RESEARCH INTERN @ Verisign, Reston, VA

May 2016 - August 2016

- Used ML (e.g. clustering) to analyze large scale HTTP traffic and protect against **DDoS** attacks.
- Acquired in-depth knowledge of **DNS** and related protocols (e.g., DNSSEC, Whois).

RESEARCH ASSISTANT @ Northeastern University, Boston, MA

September 2013 - April 2019

- Developed a large-scale and [distributed crawling](#) platform using Chromium browser instrumentation.

- Using JVM instrumentation and bytecode analysis (ASM, Soot), we built **static and dynamic analysis tools**, including a fuzzer, to detect algorithmic complexity vulnerabilities in Java, resulting in [CVE-2018-1517](#).
- Conducted a large-scale measurement of [Web Cache Deception](#) attack on popular CDNs (Akamai, Cloudflare).
- Large scale measurement of [relative path overwrite \(RPO\)](#) vulnerability using AWS and Common Crawl dataset.
- Instrument Chromium for **large-scale web security measurement** study including detection of malicious web pages/domains, ad injection by browser extensions, malvertising, and outdated JavaScript libraries.
- Conducted extensive research on the ecosystem of **online/web advertising**, its related privacy/security issues (e.g., cookie matching, re-targeted ads), security problems of **Web protocols**, and client-side technologies (e.g., JavaScript, HTML5, browser extensions).
- Built a malware analysis infrastructure based on **Cuckoo sandbox** for large-scale detection of [Ransomware](#).
- Implemented a **content distribution network (CDN)** using **DNS redirection**, user-space threads in Linux, a file system using FUSE library in Linux, and a secure chat system using built-in Java security framework.

PART-TIME SOFTWARE ENGINEER, Tehran, Iran

September 2004 - August 2013

- Developed and deployed a large-scale **SIEM** log management system using Java, **MySQL**, Hadoop, and MapReduce to efficiently collect and process logs from diverse enterprise network devices, including firewalls, IDSes/IPses, desktops, and servers.
- Researched security and performance evaluation of **shared Web hosting** solutions for **Apache web server**.
- Developed a machine learning-based [botnet](#) detection system in C++ that analyzes network communication patterns between infected hosts and command-and-control (C&C) servers. Utilizing the WEKA tool for clustering, the system identifies anomalous network behavior indicative of botnet activity, providing a novel approach to cybersecurity threat detection.
- Built a robust system capable of processing large volumes of **PCAP** network traffic and transforming them into netflow data, enabling the identification and analysis of malicious attack patterns within the network.
- Created a unified **database** schema to consolidate student and personnel information, effectively centralizing data previously spread across disparate Oracle and MS SQL Server databases within legacy systems.
- Gained a broad range of experience developing **Web applications** in diverse sets of technologies including PHP, J2EE, and ASP.NET, alongside database management using Oracle, MySQL, and MS SQL Server.

EDUCATION

Northeastern University, PhD in Computer Science
Shahid Beheshti University, MS in Computer Engineering
University of Tehran, BS in Computer Engineering

April 2019
 January 2011
 August 2008

PUBLICATIONS

- The Matter of Captchas: An Analysis of a Brittle Security Feature on the Modern Web, **WWW**, 2024
- Cached and Confused: Web Cache Deception in the Wild, **USENIX Security**, 2020
- HotFuzz: Discovering Algorithmic Denial-of-Service Vulnerabilities Through Guided MicroFuzzing, **NDSS**, 2020
- A Longitudinal Analysis of the ads.txt Standard, **ACM IMC**, 2019
- On the Effectiveness of Type-based Control Flow Integrity, **ACSAC**, 2018
- How Tracking Companies Circumvented Ad Blockers Using WebSockets, **ACM IMC**, 2018
- Large-Scale Analysis of Style Injection by Relative Path Overwrite, **WWW**, 2018
- Thou Shalt Not Depend on Me: Analysing the Use of Outdated JavaScript Libraries on the Web, **NDSS**, 2017
- Recommended For You: A First Look at Content Recommendation Networks, **ACM IMC**, 2016
- Identifying Extension-based Ad Injection via Fine-grained Web Content Provenance, **RAID**, 2016
- Tracing Information Flows Between Ad Exchanges Using Retargeted Ads, **USENIX Security**, 2016
- UNVEIL: A Large-Scale, Automated Approach to Detecting Ransomware, **USENIX Security**, 2016
- Include Me Out: In-Browser Detection of Malicious Third-Party Content Inclusions, **FC**, 2016
- Alert Correlation Algorithms: A Survey and Taxonomy, **CSS**, 2013
- A Comprehensive Approach to Abusing Locality in Shared Web Hosting Servers, **IEEE TrustCom**, 2013
- Two Novel Server-Side Attacks against Log File in Shared Web Hosting Servers, **IEEE**, 2012
- Performance Evaluation of Shared Hosting Security Methods, **IEEE TrustCom**, 2012
- An Anomaly-based Botnet Detection Approach for Identifying Stealthy Botnets, **IEEE ICCAIE**, 2011